

NAME: BIRGITHA AMSI BASHAGHE

REG NO: 24100934340073

COURSE: BETS.

TASK NO 3.

- a) How many hosts can you have in a subnet with a /26 subnet mask?

Solution.

Given.

Number of bits in subnet(n) = 26

Number of hosts = required,

from,

$$\text{Number of hosts} = 2^{(32-n)} - 2$$

$$= 2^{(32-26)} - 2$$

$$= 2^6 - 2$$

$$= 62$$

∴ There are 62 hosts in a subnet with a /26 subnet.

- (b) Discuss any four reasons why subnetting is vital?

Answer.

Four reasons for subnetting is vital.

- (i) Reduces network congestion. Subnetting improving efficiency due to smaller subnets limit broadcast domains.
- (ii) Enhanced security. Subnetting enables the creation of security boundaries, isolating sensitive areas of the network from others.

(b) (iii) Optimizes IP Address Usage. Subnetting prevents wastage by allocating addresses as needed so as to manage network easy.

(iv) Simplifies Management. Smaller subnets are easier to manage and troubleshoot than large monolithic networks.

(c). You are given the network address 192.168.1.0, the number of needed subnets 4 and the number of usable hosts 30. Determine the following:-

- i. The new subnet mask.
- ii. The total number of bits borrowed.
- iii. The 2nd subnet address range.
- iv. The subnet address for the 3rd subnet.
- v. The subnet broadcast address for the 4th subnet.
- vi. The assignable address for the 1st subnet.

Solution.

Given,

Network address = 192.168.1.0

Number of subnets needed = 4

Number of hosts needed per subnet = 30

Default subnet mask = 255.255.255.0

Then,

128	64	32		16	8	4	2	1
1	1	1		0	0	0	0	0

(c) New subnet mask 255.255.255.224
 (11111111.11111111.11111111.11100000)

Subnet	Network Prefix	Usable IP Address	Broadcast
1 st Subnet	192.168.1.0	192.168.1.1 — 192.168.1.30	192.168.1.31
2 nd Subnet	192.168.1.32	192.168.1.33 — 192.168.1.62	192.168.1.63
3 rd Subnet	192.168.1.64	192.168.1.65 — 192.168.1.94	192.168.1.95
4 th Subnet	192.168.1.96	192.168.1.97 — 192.168.1.126	192.168.1.127

(i) The new subnet mask

$$= 255.255.255.224$$

(ii) The total number of bits borrowed

$$2^n = 4$$

$$2^n = 2^2$$

$$n = 2$$

∴ The total number of bits borrowed = 2.

(iii) The 2nd subnet address range.

$$= 192.168.1.32 — 192.168.1.63.$$

(iv) The subnet address for the 3rd subnet.

$$= 192.168.1.64.$$

(v) The subnet broadcast address for the 4th subnet.

$$= 192.168.1.127.$$

(vi) The assignable address for the 1st subnet.

$$= 192.168.1.1$$